

Type Conversion In Java

Type conversion ka matlab hota ha ki hum kisi **datatype** ko ek-dusre me convert kr sakte ha. **Ex:** Group **integer** me hum **byte** ko **short, int, long** me convert kr skte ha.

Types of “Type Conversion In Java”

1. **Implicit:** Jo type conversion java khud se kr de. [Small bit datatype to large bit datatype]

Example: **byte b = 24;** $\Rightarrow$ **int i = _;** $\Rightarrow$ **i = b;** $\Rightarrow$ **system.out.println (i);** = 24 print ho jyga, and this conversion is known as **WIDENING CONVERSION**.

$\therefore$, **RULE FOR IMPLICIT CONVERSION** $\Rightarrow$

Destination datatype should be wider (more bits) then source datatype.

2. **Explicit:** Jo type conversion hume manually krna pde **CASTING** ke through. [large datatype to small datatype].

Example: **int ii = 300;** $\Rightarrow$ **byte b = _;** $\Rightarrow$ **b= ii;** ❌

Int ii = 300; $\Rightarrow$ **byte b = _;** $\Rightarrow$ **b = (byte) ii;** ✅ $\Rightarrow$ **system.out.println (b);** = kuche ye process hoga. (Step 1) ii = 300, binary me convert ho jyga = {100101100}

Pr 300 int value ha to iski range 32bits ha to iski age ki values me 0's lg jyge.

8 bits	8 bits	8 bits	8 bits
00000000	00000000	00000001	00101100

Ab byte be ki value last wala **cell** hoga kuki byte ki value 8 bits ki hoti ha.

$\therefore$ **byte b = 00101100** $\Rightarrow$ **44 ans,**

And this conversion is known as **NARROWING CONVERSION**. `int ii = 300;`

```
int ii = 300;
byte bb;
bb = (byte)ii;
System.out.println("case 2 of explicit [byte to int]-->" + bb);
```

Result: case 2 of explicit [byte to int]-->44

Automatic Type Promotions Conversion

Isme java compiler automatically small datatype ko bade datatype me convert kr deta ha, taki data loss na ho.

Types of Automatic Type Promotion Conversion

1. Byte, short and char values are promoted to int.
2. If 1 operand is long, then whole expression will become long.
3. If 1 operand is float, then whole expression will become float.
4. If 1 operand is double, then whole expression will become double.

```
//Type promotions
// agar ab hm asa kre byte b = 50; --> b = b*2; to kya ye possible ha?
//or isme type mismatch wala error ku aa ra ha int to byte conversion wala iska solu notes
me ha.

byte bpp = 50;
bpp = (byte) (bpp*2); // ab result 100 ana chahiye
System.out.println("Automatic Type promotion-->" + bpp);

// Applying all four rules of Automatic Type Promotions here is code below
byte bbb = 42;
char ccc = 'a';
```

```

short sss = 1024;
int iii = 50000;
float fff= 5.67f;
double ddd = .1234;
// yha pr dekho sabhi datatypes me double datatype sabse bda ha to sara result double me hi
calculate hoga.
// isi liya niche jo bhi calculatons ha usme hum double datatype ka use kr rhe ha.
double result = (fff * bbb) + (iii / ccc) - (ddd * sss);
/*ab yha pr bhi har category me result bde datatype me hi niklega.
1) fff * bbb = float
2) iii / ccc = int
3) ddd * sss = double
therefore, float + int = float - double = double */
System.out.println((fff*bbb) + " + " + (iii / ccc) + " - " + (ddd * sss));
System.out.println("result="+ result);

```

Result automatic Type promotion-->100

238.14 + 515 - 126.3616

result=626.7784146484375